DETAILED IMPLEMENTATION PHASES

Souqly

AI-Powered Small Business Marketplace · Egypt
Arabic & English · ASP.NET Core MVC

Version	2.1 — Corrected (review pass)
Based on	SRS Souqly v2.1 (single source of truth)
Style	Tasks + acceptance; rules live in SRS / docs/rules.md
Rule	Finish each phase before the next; unit tests with each module
AI Timing	After Phases 2–4 core is stable

This document breaks **Souqly** into 8 detailed phases. Each phase lists concrete tasks, important details from the SRS, and a clear acceptance check. Core marketplace first — AI features after the foundation is stable.

1. Phase Overview

#	Phase	Focus	Key Deliverable
1	**Requirements & Design**	MVP, ERD, rules, architecture	SRS, ERD, project structure
2	**Foundation & Auth**	Solution, Identity, roles, profiles	Users register, log in, correct access
3	**Stores, Products & Discovery**	Storefronts, products, search, admin	Sellers publish; customers discover
4	**Orders, Payments & Delivery**	Cart, multi-seller checkout, COD, delivery	Customer places order and tracks it
5	**AI Integration**	Listing, image analysis, pricing, assistant	Working AI features (after core is stable)
6	**Finance, Refunds & Notifications**	Commission, refunds, notifications, reports	Full financial and notification flows
7	**Quality & Security**	Tests, security, logging, edge cases	Reliable app, not just a happy-path demo
8	**Deployment & Presentation**	Deploy, demo data, docs, presentation	Live demonstrable project

Key principle: *Phase 1 = design only. Core build =* ***Phases 2–4*** *(stable before AI in Phase 5). Write unit tests with each module; Phase 7 fills gaps + security/integration. Rules: SRS v2.1 + docs/rules.md (do not re-copy full rules here).*

2. Phase Details

PHASE 1 — Requirements and System Design

Goal: Agree how Souqly works before writing production code.

1.1 Define MVP and User Stories

- Extract MVP scope from SRS (Core marketplace first; AI after core is stable).
- Write user stories for: Guest, Customer, Seller, Administrator, Delivery Provider.
- Prioritize stories that unlock the first end-to-end path (browse → cart → order).
- List explicit out-of-scope items: food features, guest checkout, GPS tracking, auto split payouts, post-delivery RMA/returns workflow.

1.2 ERD & main flows

- Main flows (text is enough): seller application → approval → publish product → discovery → multi-seller checkout → delivery → refund.
- ERD covering: Users, Roles, SellerProfiles, Addresses, Categories, Products, ProductImages, ProductAttributes, CartItems, ParentOrders, SellerOrders, OrderItems, Payments, Refunds, Shipments, DeliveryProviders, Governorates, SellerGovernorates, DeliveryFees, Commissions, Reviews, Notifications, IdempotencyKeys, SystemSettings, AuditLogs.
- Document key relationships (especially ParentOrder → SellerOrders → Shipments).

1.3 Business Rules (from SRS Core Rules)

- Product rules: required fields, non-negative price/stock, at least one language, historical snapshot on order.
- Image rules: up to 5 images/product, max 5 MB, JPG/PNG/WebP.
- Order rules: multi-seller cart → one parent order + separate seller orders; unique order reference; validation at checkout; idempotency.
- Status model: Seller Order (Pending→Accepted→Processing→Shipped→Completed/Cancelled); Shipment (Unassigned→Assigned→Picked Up→In Transit→Delivered/Failed); Parent derived from seller orders.
- Seller accept/reject: accept or reject full seller order (no partial line-item accept in MVP); optional reject reason.
- Seller timeout: auto-cancel Pending seller orders after configurable timeout (default 48 hours).
- Customer cancel: one-click while Pending; request after Accept; no normal cancel after Shipped/Delivered.
- Payment rules: COD tracking; provider marks COD collected on delivery; online confirmed only by gateway; no raw card storage; no guest checkout.
- Delivery rules: fixed fee by governorate; seller coverage check; status transitions.
- Stock decreases at order placement; restore on Pending cancel/reject/timeout. Money: EGP decimal; net = subtotal + fee – commission – refunds (see SRS example).
- Commission: (product subtotal + delivery fee) × rate; no discounts in MVP; rate change does not alter confirmed orders.
- Cancellation: Pending auto; after Accept request flow; full seller-order refunds only (partial qty = Stretch).
- Online: AwaitingPayment + 30 min expire; COD first; FakeGateway until sandbox. Only Admin assigns shipments.
- In-app notifications start with orders (Phase 4); email/SMS polish later. Tests written per module; Phase 7 fills gaps.
- Follow SRS v2.1 + docs/rules.md — do not re-implement conflicting local rules.

1.4 Architecture & Team

- Confirm stack: ASP.NET Core MVC, EF Core, SQL Server, ASP.NET Core Identity.
- Define layered structure: Controllers / Services / Repositories (or equivalent).
- List external services: Payment gateway, AI provider, Email, SMS.
- Assign responsibilities and decide bilingual approach (resource files / localization).
- Create initial solution structure and empty projects.

Deliverables: Final SRS, ERD, agreed business rules (docs/rules.md), initial MVC project structure.

Acceptance: Team agrees on MVP scope, data model, and rules. No major feature coding starts yet.

PHASE 2 — Foundation and Authentication

Goal: Secure base so every role can register/login and only see what they are allowed to see.

2.1 Solution & Database Setup

- Create ASP.NET Core MVC solution and projects for Souqly.
- Configure EF Core + SQL Server connection (secrets outside source code).
- Initial migrations for Identity tables and custom profile fields.
- Seed roles: Customer, Seller, Administrator, DeliveryProvider.

2.2 Identity, Roles & Authorization

- ASP.NET Core Identity: registration, login, logout, password recovery.
- Role-based authorization policies for each role.
- Prevent access to resources belonging to other users (ownership checks).
- Admin can suspend and reactivate accounts; suspended users cannot perform restricted actions.
- Account verification flow (email confirmation recommended).
- No guest checkout — user must be authenticated to place an order.

2.3 Profiles & Seller Application

- Customer profile: manage basic info + multiple delivery addresses.
- Seller registration application form (business name, contact, required fields).
- Application status: Pending → Approved / Rejected.
- Unapproved sellers cannot publish products or accept orders.
- Delivery Provider account creation (admin-managed).

2.4 Bilingual & Layout Foundation

- Localization setup for Arabic (RTL) and English (LTR).
- Language switcher.
- Responsive layout skeleton (desktop, tablet, mobile).
- Shared layout, navigation, and basic error pages.

Deliverable: Users can register, log in, and access only the areas allowed by their role.

Acceptance: (1) Customer and Seller can register and log in. (2) Role boundaries enforced. (3) Arabic RTL and English LTR switch correctly. (4) Passwords hashed; secrets not in source. (5) No guest checkout.

PHASE 3 — Stores, Products, and Discovery

Goal: First marketplace milestone — sellers publish products; customers can find them.

3.1 Seller Approval & Storefront

- Admin review queue for seller applications (approve / reject / suspend).
- On approval: seller can create and manage storefront.
- Storefront shows seller info, available products, and ratings area.
- Seller defines governorates they serve (delivery coverage).
- Sellers manage only their own storefront, products, and later their orders.

3.2 Categories & Products

- Admin manages categories and subcategories.
- Product fields: name, description, category, price, images, stock/availability, attributes.
- At least one language (Arabic or English) required; both recommended.
- Image upload: up to 5 images per product, max 5 MB each, JPG/PNG/WebP only.
- Stock tracking; prevent negative stock and overselling.
- Publish / unpublish / edit / remove.
- Validation of required fields and price before publication.
- Admin can moderate, hide, or remove listings that violate policy.
- When a product is ordered, snapshot name, price, quantity into the order item.

3.3 Discovery & Search

- Public browse of storefronts and published products.
- Keyword search + filters: category, price range, location/governorate, delivery availability, attributes.
- Sort by price and relevance + pagination. Relevance in MVP = text match quality + recency.
- Product detail page: seller, price, availability, images, description, delivery info.
- Exclude unpublished, suspended, or ineligible listings.

Deliverable: Sellers can publish valid products; customers can search, filter, and view them.

Acceptance: (1) Seller cannot publish before admin approval. (2) Approved seller manages only own listings. (3) Search/filter returns only eligible real products. (4) Image limits enforced. (5) Historical order data stays correct after product edit.

PHASE 4 — Orders, Payments, and Delivery

Goal: Complete purchase path — multi-seller cart, checkout, payment, delivery tracking.

4.1 Cart, Checkout & Status Model

- Add products from multiple sellers to one cart; update quantities / remove items.
- At checkout validate: current price, stock, seller eligibility, destination coverage.
- Calculate product subtotals, delivery fees per seller shipment, and grand total.
- Create one Parent Order + separate Seller Orders with unique human-readable order reference numbers.
- Display subtotal and delivery fee for each seller shipment before confirmation.
- Recalculate when cart or destination changes.
- Idempotency: prevent duplicate orders from double-submit / retries.
- Implement status model: Seller Order Pending→Accepted→Processing→Shipped→Completed/Cancelled; Shipment Unassigned→Assigned→Picked Up→In Transit→Delivered/Failed; Parent status derived from seller orders.
- Seller Accept / Reject full seller order (optional reject reason). No partial line-item accept in MVP.
- Auto-cancel Pending seller orders after configurable timeout (default 48 hours).
- Customer one-click cancel while Pending; after Accept, cancellation is a request (reviewed by seller/admin).
- No normal customer cancel after Shipped/Delivered (returns workflow is out of MVP).
- Support partial fulfillment (each seller order independent).
- Customer views parent order and status of each seller order; each seller sees only their own.

4.2 Payments

- Cash on Delivery (COD): record amount due and collection status.
- On successful COD delivery, delivery provider marks the amount as collected; admin can correct if needed.
- Online payment via integrated gateway (start with test/sandbox credentials).
- Record: method, amount, transaction reference, status.
- Confirm online payment only from trusted gateway callback — not browser alone.
- Prevent duplicate payment processing. Never store raw card details.

- Maintain financial records per seller order inside a multi-seller checkout.

4.3 Delivery

- Admin configures fixed delivery fees by destination governorate.
- Block checkout if no fee is configured for the chosen destination.
- Validate that every seller in the cart serves the destination.
- Admin creates/manages Delivery Provider accounts.
- Administrator assigns each seller shipment to a provider (Unassigned → Assigned).
- Provider sees only assigned shipments and needed delivery info.
- Status flow: Assigned → Picked Up → In Transit → Delivered / Failed.
- Record failure reasons. Customer can track progress of each shipment.
- Multi-seller orders have separate shipments, assignments, and statuses.
- GPS tracking and auto-allocation are future (not this phase).

4.4 Reviews & simple in-app notifications

- Reviews only when seller order is Completed (Delivered); one review per product per order; admin can hide (no report flow).
- Simple in-app notifications now: order placed, accept/reject, shipment status (table + bell). Email/SMS later in Phase 6.
- Work Phase 4 as slices if needed: 4A cart/checkout, 4B seller orders/timeout, 4C payments (COD first + FakeGateway), 4D delivery, 4E reviews/notifications.

Deliverable: Multi-seller purchase works (COD path solid); accept/reject/timeout/cancel; delivery trackable; basic in-app notifications.

Acceptance: (1) Multi-seller checkout creates parent + seller orders with references. (2) Totals reconcile. (3) Seller can accept/reject; timeout auto-cancels Pending orders. (4) Customer one-click cancel works while Pending. (5) COD and online test flows record correct states; provider can mark COD collected. (6) Correct governorate fee applied. (7) Customer sees statuses; provider updates only own shipments.

PHASE 5 — AI Integration (After Core is Stable)

Goal: Integrate AI as assistive tools. Start only after Phases 2–4 are working and stable.

AI is part of MVP but scheduled after the core marketplace. Follow SRS Core Rules 3.1 (assistive only).

5.1 Shared AI Infrastructure

- Create IAIService interface and concrete implementation.
- Configurable API key, timeouts, usage limits, and cost controls (outside source code).
- Structured output parsing and validation before any save.
- Graceful error handling: timeouts, invalid responses, provider downtime.
- AI failures must not lose the user's original input or corrupt data.
- Log AI calls for debugging and cost monitoring (no sensitive data in logs).

5.2 AI Product Listing Assistant (first feature)

- Seller submits basic product info + category.
- AI suggests: title, description, tags, searchable attributes (Arabic & English).
- UI shows suggestions; seller can edit, accept, or reject each part.
- Nothing is published without explicit seller confirmation.

5.3 AI Product Image Analysis

- Seller uploads product image(s).
- AI suggests visible attributes, styles, colors, descriptive details.
- Show image + suggestions side by side; seller chooses what to apply.
- Present as suggestions only — never as verified facts.

5.4 AI Pricing Assistance

- Seller can provide cost inputs (materials, labor, packaging).
- System generates price guidance; display assumptions where available.
- AI never changes a published price without explicit seller confirmation.
- Seller remains fully responsible for the final price.

5.5 AI Conversational Shopping Assistant

- Customer describes need in natural language (Arabic or English).
- AI extracts criteria; system queries the real live catalog.
- Results respect current price, stock, seller coverage, and delivery eligibility.
- If nothing matches, explain clearly. Assistant cannot place orders or change data.

Deliverable: All four AI features work inside the app and follow the assistive-only rules.

Acceptance: (1) Seller generates bilingual listing, edits, and explicitly approves. (2) Image analysis shows suggestions only. (3) Pricing guidance does not auto-change price. (4) Shopping assistant returns only real eligible catalog products. (5) AI failures do not corrupt data.

PHASE 6 — Finance, Refunds, and Notifications

Goal: Complete money flows, refunds, and reliable notifications on top of working orders.

6.1 Platform Commission & Admin Config

- Admin configures: platform-wide commission rate, delivery fees by governorate, seller acceptance timeout, AI limits, notification channels.
- Formula: Commission basis = product subtotal + seller shipment delivery fee (no discounts in MVP).
- Commission = basis × rate. Calculated and stored per seller order.
- Sellers can view: product amounts, delivery fees, commission deductions, refunds, net earnings.
- Rate change must not alter already confirmed orders.
- Commission kept separate from delivery-provider costs and seller payouts.
- Provider cost is admin reporting only — no automated provider settlement in MVP.
- MVP records net earnings only — automated split payouts are future.

6.2 Cancellations & Refunds

- Eligible cancellation before seller acceptance → automatic (including timeout auto-cancel).
- After acceptance → policy checks; seller reviews ordinary requests; admin handles disputes.
- Each seller order has independent cancellation/refund outcome.
- MVP: full seller-order refund only (partial quantity = Stretch).
- Online payments: initiate/record refund through payment gateway.

- COD: record “refund payable” → admin approves electronic refund → mark settled only after confirmation.
- Prevent duplicate refunds and refunds exceeding remaining eligible amount.
- Stock restoration default: restore on pre-acceptance cancellations and admin-approved restock cases. Do not restore on delivered + customer-remorse unless policy explicitly allows.
- Returns after delivery (RMA / return shipping) are out of MVP — post-delivery issues are manual admin exceptions only.
- Preserve original transaction + auditable history of all adjustments.

6.3 Notifications (matrix)

- Channels: in-app, email, SMS (configured providers).
- Order placed → Customer, each affected Seller.
- Seller order accepted / rejected → Customer, Seller.
- Seller order auto-cancelled (timeout) → Customer, Seller.
- Shipment assigned → Delivery Provider, Seller.
- Shipment status updates (picked up, in transit, delivered, failed) → Customer, Seller.
- COD marked collected → Admin (and Seller summary as applicable).
- Cancellation request / outcome → Customer, Seller, Admin (for disputes).
- Refund requested / approved / completed / failed → Customer, Seller, Admin.
- Notification failure must not reverse the business transaction.
- Never include passwords, card details, or unnecessary sensitive data.

6.4 Reporting & Policy

- Basic admin operational counts and transaction summaries.
- Seller basic sales and commission summaries.
- Admin can manage marketplace rules and prohibited-product policies.
- AI suggestions never override policies.

Deliverable: Commission, full-seller-order refunds (online + COD), expanded notifications, admin config work correctly.

Acceptance: (1) Commission + net earnings match SRS example. (2) Pre-accept/timeout cancel automatic. (3) Full seller-order refund paths without over-refund. (4) Notifications reach correct users; failures do not break orders. (5) No RMA workflow.

PHASE 7 — Quality and Security

Goal: Turn a working demo into a reliable, tested application.

7.1 Automated Testing

- Unit tests for: order total calculation, commission formula, stock changes, refund limits.
- Integration tests for: checkout flow, payment status handling, cancellation rules.
- Tests for seller accept/reject, Pending timeout auto-cancel, customer one-click cancel while Pending.
- Tests for COD collected by provider and full seller-order refund limits.
- Authorization tests: each role cannot access unauthorized resources.
- Ownership tests: seller A cannot see seller B's products/orders.
- Idempotency tests: repeated checkout / payment / refund requests do not create duplicates.
- Image upload validation tests (count, size, format).
- AI failure tests: invalid response, timeout — data stays clean.

7.2 Security Hardening

- Review authentication and session configuration.
- Input validation and protection against common web vulnerabilities.
- Secure file uploads (type, size, storage location) matching SRS image limits.
- Basic rate limiting on registration, login, search, and AI calls.
- All secrets outside source code. Passwords securely hashed.
- Principle of least privilege for database and service accounts.

7.3 Reliability & Observability

- Structured logging for important business events and errors.
- Global exception handling that does not leak sensitive details.
- Pagination everywhere lists can grow.
- Database backup and restoration procedure written and tested once.
- Notification retry strategy without repeating business actions.

7.4 Accessibility & Polish

- Clear labels, understandable validation errors.
- Keyboard navigation for main flows. Sufficient color contrast.
- Final pass on Arabic RTL and English LTR layouts.

Deliverable: A tested, reliable application — not only a happy-path demo.

Acceptance: (1) Critical automated tests pass. (2) Authorization and ownership boundaries hold. (3) Duplicate and failure scenarios do not corrupt records. (4) Secrets and uploads are secure. (5) Backup procedure exists.

PHASE 8 — Deployment and Presentation

Goal: Put a live, demonstrable Souqly system in front of evaluators with a clear AI story.

8.1 Deployment

- Deploy application to target hosting environment.
- Configure production (or staging) database and run migrations.
- Set all environment variables and secrets correctly.
- Configure payment gateway test credentials, AI, email, and SMS providers.
- Verify bilingual and responsive behavior on deployed site.

8.2 Demo Data & Accounts

- Create test accounts: at least one Customer, two Sellers, one Admin, one Delivery Provider.
- Seed realistic products across categories and governorates.
- Prepare a multi-seller cart scenario and sample orders in different statuses.
- Have sample AI-generated listings ready to show edit/approve flow.

8.3 Documentation

- Setup / installation instructions.
- Environment variables and configuration reference.
- Short notes for payment and AI integrations.
- Known limitations and future enhancement list.

8.4 Presentation

- Prepare a clear walkthrough of a complete customer journey on Souqly:
- Seller applies → Admin approves → Seller creates product (with AI listing help) → Customer discovers product → multi-seller cart → checkout (COD or online test) → Order created → Seller accepts → delivery assigned → status updates → provider marks COD collected → notifications → Seller sees earnings and commission.
- Also demo: customer cancel while Pending; seller reject (optional); bilingual UI.
- Explicitly highlight the AI contribution and the assistive-only design.
- Be ready to show failure handling, role separation, and timeout/auto-cancel if asked.

Deliverable: Live demonstrable Souqly project with setup docs and a clear explanation of its AI contribution.

Acceptance: Full end-to-end demo runs successfully; AI value is explained; documentation allows another person to run the system.

3. Future Enhancements (After MVP)

Not required to finish the project. Start only after Phases 1–8 are complete and stable.

- AI background removal and professional product-scene generation
- Image-based visual product search
- Voice-based product listing creation
- Automatic delivery-provider assignment and route optimization
- GPS-based live delivery tracking
- Automated marketplace split payments and seller payouts
- Advanced analytics, forecasting, personalized recommendations
- Weight/dimension-based delivery pricing
- Advanced dispute management and automated returns

4. Rules That Apply Across All Phases

- Finish and agree each phase before starting the next.
- Core marketplace (Phases 2–4) must be stable before starting AI (Phase 5).
- All financial calculations use deterministic application logic — never AI.
- AI is always assistive; seller or customer stays in control.
- Confirmed orders keep original terms; later changes are recorded as adjustments.
- Seller accepts/rejects full seller order only; Pending timeout auto-cancels (default 48h).
- Customer one-click cancel only while Pending; returns after delivery are out of MVP.
- Notification failures never reverse a successful business transaction.
- No guest checkout. No food-specific features. No post-delivery RMA workflow in MVP.
- Security and authorization enforced from Phase 2 onward; basic rate limiting included.
- Expand automated tests as each phase is completed.
- Never store raw payment-card details.
- Policy or rate changes must not silently alter existing order financials.
- Follow SRS Core Rules as the single source of truth when phases and SRS wording differ.

— End of Detailed Implementation Phases —
Souqly · Phases Version 2.1 · Based on SRS v2.1 · Core = Phases 2–4 · AI after stable